GALA: Graph-Augmented LLM Agents for Root Cause Analysis and Incident Response in Microservices

Supplementary Technical Report

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1 Extended Evaluation on AegisLab

Setup. AegisLab is a fault propagation-aware benchmark for microservice root cause analysis spanning 25 distinct fault types [1], more than three times the six fault types shared by OnlineBoutique (OB) and TrainTicket (TT). We evaluate 7 top-performing baselines, comprising of 5 non-LLM and 2 LLM-based baselines, on 100 AegisLab fault cases. Baselines use their published configurations, and all LLM-based methods share the backbone used in the main paper.

Table 1: Root cause ranking on 100 AegisLab fault cases. Best in bold, second best underlined. Improvement is the relative gain over the second-best entry in each column.

Type	Method	AC@1 (%)	AC@3 (%)	MRR
Non-LLM	CIRCA	9.0	21.0	0.202
	RCD	3.0	11.0	0.069
	HeMiRCA	24.0	37.0	0.355
	BARO	10.0	22.0	0.227
	Nezha	18.0	37.0	0.298
LLM-based	mABC	15.0	21.0	0.175
	Flow-of-Action	<u>28.0</u>	<u>52.0</u>	<u>0.381</u>
	GALA	**30.0**	**62.0**	**0.477**
Improvement over second best		+7%	+19%	+25%

Results. GALA attains the best value on all three ranking metrics (Table 1). Its margin over Flow-of-Action, the strongest baseline on this benchmark, is +0.096 MRR (+25% relative) and +10 percentage points AC@3 (+19% relative). We note explicitly that the AC@1 margin is small: 30 versus 28 correctly ranked incidents out of 100, a difference of two cases that is not statistically significant. The AegisLab claim therefore rests on MRR and AC@3, where the margins are substantially larger; we do not claim a significant AC@1 advantage on this benchmark. Among the non-LLM baselines, HeMiRCA is strongest (0.355 MRR, 24.0% AC@1), followed by Nezha (0.298 MRR), while CIRCA and RCD—the two leaders on OB—place sixth and eighth.

Difficulty. AegisLab is markedly harder than OB or TT for every method evaluated. No method exceeds 30% AC@1, and the best MRR attained on the benchmark (0.477) falls well below GALA's own MRR on OB (0.854) and TT (0.801). The broader fault taxonomy and the explicit modeling of fault propagation leave substantial headroom, and we do not present these results as a solved benchmark. The claim is comparative: GALA's ranking advantage is preserved, and in relative terms widens, as fault diversity increases.

References

[1] Aoyang Fang, Songhan Zhang, Yifan Yang, Haotong Wu, Junjielong Xu, Xuyang Wang, Rui Wang, Manyi Wang, Qisheng Lu, and Pinjia He. 2026. Rethinking the Evaluation of Microservice RCA with a Fault Propagation-Aware Benchmark. *Proceedings of the ACM on Software Engineering* FSE (2026).